TEXORA · LECTURE SHEET

Project Plan for a New Garments Factory

From product choice to payback: sizing capacity from SMV, listing the machinery, planning the floor and the manpower, and assembling the investment — the course's calculations combined into one project.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 10 · SPECIALIZATION — APPAREL MANUFACTURING

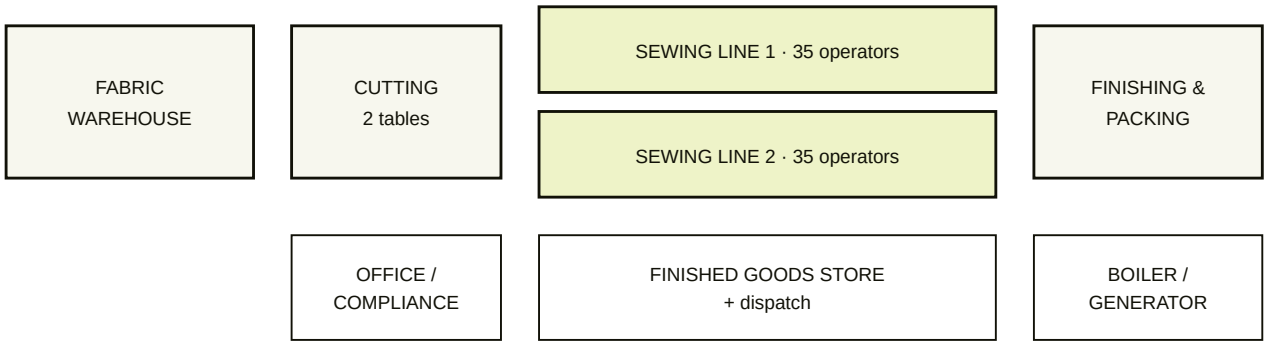
1. The sequence of a factory project

A garments factory is planned backwards from its market. The sequence is fixed: choose the **product** (here, basic woven shirts — the product whose calculations fill this course); fix the **capacity** in pieces per day from a realistic order book; derive **machines and manpower** from capacity and SMV (Topics 7–8); lay out the **floor** in material-flow order (Topic 1's machinery map, Topic 8's floor optimisation); then price the whole project and test it against the **payback** its prices can earn (Topic 6). Skipping the order of these steps — buying machines before capacity, or a building before the floor plan — is how factories are born unbalanced.

2. Capacity plan

WORKED EXAMPLE — SIZING THE PROJECT

Product: woven shirts, SMV 20 min. Line design: **2 lines × 35 operators**.
Output per line at 70% working efficiency = (35 × 480 × 0.70) ÷ 20 = **588 pcs/day**.
Factory output = 2 × 588 = **1,176 pcs/day**; at 26 working days, **30,576 pcs/month** — roughly two and a half course-size orders (12,000 pcs) per month, a credible book for a new factory with two anchor buyers.



Floor blocks in material-flow order: fabric in at the left, finished cartons out at the right

Figure 1 — Schematic floor plan of the project factory (20,000 sq ft built-up). Sewing occupies the centre because every other department feeds it or is fed by it.

3. Machinery and investment

ITEM	QUANTITY / BASIS	US$
Single-needle lockstitch machines	70 × $400 (one per sewing operator)	28,000
Overlock machines	10 × $650	6,500
Buttonhole machines	2 × $3,500	7,000
Button-sewing machines	2 × $2,800	5,600
Feed-off-the-arm machines	2 × $1,200	2,400
Cutting section (straight knife, band knife, tables, spreader)	Set	8,500
Fusing press	1	4,200
Pressing & finishing (boiler, vacuum tables, form finisher)	Set	6,800
Generator, compressor, needle detectors, misc.	Set	15,000
Machinery subtotal		**84,000**
Building, 20,000 sq ft built-up	× $28/sq ft	560,000
Electrical installation, ETP share, furniture, pre-operating costs	Lump sum	45,000
Total project investment		**689,000**

4. Manpower plan

DEPARTMENT	STAFF	NOTE
Sewing operators	70	2 lines × 35 (capacity plan)
Cutting	8	Cutters, spreaders, layers, bundling
Finishing & packing	12	Thread trimming, pressing, folding, polybagging, cartoning
Quality	6	In-line, end-of-line and audit inspectors (Topic 9)
Supervision, IE, merchandising, admin, compliance	10	Line chiefs, IE officer, store and office staff
Total	**106**	

5. Payback test

WORKED EXAMPLE — SIMPLE PAYBACK

Monthly revenue at the Topic 6 FOB of $4.70: 30,576 × 4.70 = **$143,707**.
At a planned net margin of 8% of revenue: monthly net profit = 143,707 × 0.08 = **$11,497**.
Simple payback = 689,000 ÷ 11,497 ≈ **60 months (5 years)**.
The building is 81% of the investment — which is why so many real factories of this size rent their premises: renting removes most of the $560,000 from the project cost and shortens payback drastically, at the price of monthly rent. The plan must state which route it takes; the arithmetic differs, the method does not.

Glossary

TERM	MEANING
Project plan	The integrated capacity, machinery, floor, manpower and investment design of a new factory.
Simple payback	Total investment ÷ periodic net profit: the time for profits to repay the investment, ignoring interest and discounting.
Anchor buyer	A committed customer whose recurring orders underwrite a new factory's capacity.

Pre-operating costs	One-off expenses before production starts: licences, installation, recruitment, trial runs.

Key points and summary

- Plan backwards: product → capacity → machines/manpower → floor → investment → payback.
- Capacity: 2 lines × 35 operators at SMV 20 and 70% efficiency give 1,176 pcs/day, 30,576/month.
- Investment: machinery $84,000; building $560,000; other $45,000; total $689,000 for 106 staff.
- Payback at an 8% net margin is about 60 months — dominated by the building, which is the case for the rent-versus-build decision.

Practice

Answer in your own words; brief model answers follow.

1. The factory adds a third identical line. Recalculate daily and monthly output, and the lockstitch machine cost.
2. Working efficiency reaches only 60% in the first year. Recalculate monthly output and the simple payback at the same 8% margin on revenue.
3. Why must capacity be fixed before the machinery list, rather than buying machines and seeing what they produce?
4. State two items in the investment table whose cost would change if the product changed from woven shirts to knitted T-shirts.

Answer key

1. Per line 588/day → 3 lines = 1,764 pcs/day; monthly = 1,764 × 26 = 45,864 pcs. Lockstitch: 105 machines × $400 = $42,000.
2. Output per line = 35 × 480 × 0.60 ÷ 20 = 504/day; factory = 1,008/day; monthly = 26,208 pcs. Revenue = 26,208 × 4.70 = $123,178; profit = × 0.08 = $9,854; payback = 689,000 ÷ 9,854 ≈ 70 months.
3. Machine types, counts and the floor all derive from the output target and SMV. Buying first fixes capacity by accident — usually unbalanced, with bottlenecks in one section and idle machines in another.
4. Examples: buttonhole/button-sew machines drop out (knit polos use different closures); overlock count rises (knit construction is overlock-based); fusing press demand falls; cutting and finishing sets change. (Any two well-reasoned items.)